# Context-Parallel Mamba2: Scaling to 1M+ Tokens

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## Mamba2: A Linear Attention Layer

The quadratic $\mathcal{O}(\texttt{seqlen}^2)$ scaling of Transformers attention has increasingly become a bottleneck as context lengths have ballooned with the advent of reasoning models and rising prevalence of modalities such as video and audio.

| Architecture | **Prefill/Train** | **Decode** |
|---|---|---|
| Attention | $\mathcal{O}(\texttt{seqlen}^2)$ | $\mathcal{O}(\texttt{seqlen})$ |
| Mamba2 | $\mathcal{O}(\texttt{seqlen})$ | $\mathcal{O}(1)$ |

Mamba2 [1] is an alternative information propagation algorithm belonging to the steadily growing class of *linear* $\mathcal{O}(\texttt{seqlen})$ attention mechanisms [2]. Its GPU-aware design enables hardware utilization comparable to attention during training, and reduces the decoding time and cache-space from $\mathcal{O}(\texttt{seqlen})$ to $\mathcal{O}(1)$. Models utilizing Mamba2 layers include [3], [4], [5], [6].

## Mamba2 Architecture (Simplified)

Mamba2 relies on two central mechanisms for propagating information:
1. Short 1D causal convolutions [7]
2. A gated recursion relation

Giving inputs $x_{sd} \in \mathbb{R}^{\texttt{seqlen} \times \texttt{d\_model}}$, the Mamba2 outputs are schematically

$$z_{sd} \sim \texttt{gated\_recursion}(\texttt{causal\_conv1d}(x_{sd}))$$

Both components require adaptation in the CP implementation.

### Causal Convolutions

The causal convolution [7] is a depthwise, 1D convolution along the sequence dimension with a short filter, typically of width $K = 4$ :

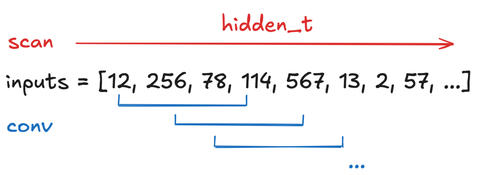
$$z_{sd} = \sum_{k=0}^{K} W_{kd} x_{(s-k)d} \ .$$

### Gated Recursion Relations

The central elements in the Mamba2 recursion relation are of the form:

$$z_{sd} = e^{-A_s} z_{(s-1)d} + \Delta_s x_{sd}$$

where the data-dependent $A_s, \Delta_s \geq 0$ control the deletion and addition of information to the state $z_{sd}$. While the complete tensor $z_{sd}$ can be constructed in $\mathcal{O}(\texttt{seqlen})$ by solving the recursion in the naive manner, such an approach is suboptimal in practice as it cannot leverage GPU tensor cores. For this reason, the recursion is solved using a chunked, matmul-based strategy which has inferior theoretical scaling, but superior in-practice wall time [1].




## Context Parallelism

Leveraging Mamba2's long-context scaling advantages requires long-context training which poses engineering challenges as $\texttt{activation\_mem} \propto \texttt{seqlen}$ .

Context-parallelism (CP), in which sequences are sharded along the sequence dimension, is a natural and scalable approach to long-sequence training.

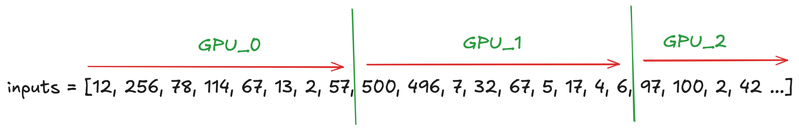



Figure 2: Context Parallel Sharding

## CP Convolutions

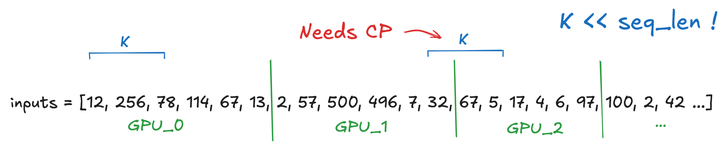



Figure 3: Context Parallel Causal Convolution

Only minimal modifications required for CP causal convolutions. Each GPU handles $C = \texttt{seqlen} \ / \ \texttt{cp\_degree}$ tokens, and only the outputs for the first $K - 1 \ll C$ tokens require communication for computational correctness.

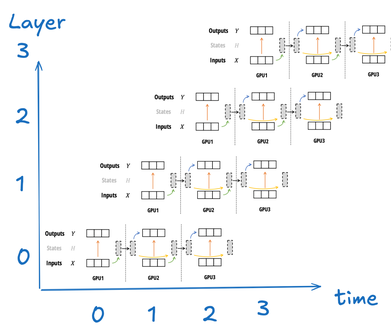
An efficient algorithm is as follows:
1. CP rank `r` asynchronously passes its final $K - 1$ tokens to rank `r+1`
2. Concurrently, ranks convolve their local tokens: `z = causal_conv1d(x)`
3. The leading outputs are corrected via the received tokens:
   `z[:K-1] = causal_conv1d(cat(x_recv, x[:K-1]))`

## CP Recursion: State Passing

A straightforward CP implementation of Mamba2's recursion step comes from passing state across CP boundaries. This proceeds as follows:

1. Rank $r$ solves the recursion locally and passes `z[-1]` to rank `r + 1`.
2. Rank $r + 1$ waits on the passed state, then solves recursion locally.

The causal Mamba2 dependencies result in $\mathcal{O}(\texttt{seqlen})$ exposed communication. However, pipelining can be use to amortize this cost [1].




## CP Recursion: Compute-Then-Correct

An alternative implementation which can have better strong-scaling properties is as follows:

1. Every rank computes $\Sigma_r = \sum_c A_{rc}$ and async sends to ranks $r' > r$
2. Every rank solves their local recursion, producing (incorrect) states $y_{rd}^{\text{final}}$
3. Rank $r$ sends $y_{rd}^{\text{final}}$ to CP ranks $r' > r$
4. Rank $r$ corrects its initial state: $x_{rd}^{\text{initial}} = \sum_{r'<r} \exp\left(\sum_{r''=r+1}^{r-1} \Sigma_{r''}\right) y_{r'd}^{\text{final}}$
5. Every rank re-solves the recursion relation with its corrected initial states

This strategy requires $\mathcal{O}(\texttt{seqlen} \ / \ \texttt{cp\_degree})$ additional compute and has $\mathcal{O}(\texttt{cp\_degree})$ exposed communication.

## Scaling

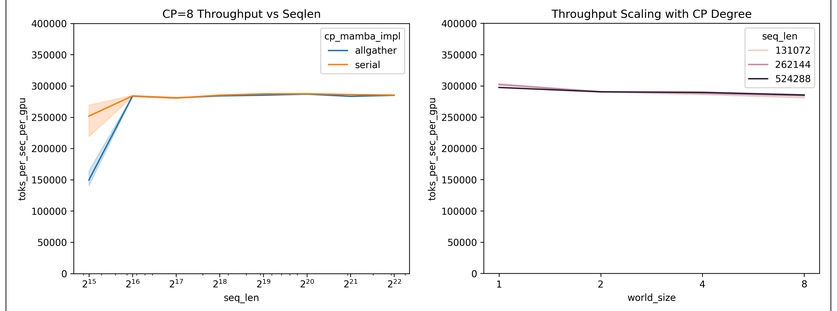



Figure 5:

Scaling performance for a stack of eight `d_model=4906` Mamba2 layers on H100 GPUs. Left: throughput vs `seqlen` at `cp_degree=8`. Per-GPU performance increases with `seqlen` since compute $\sim \mathcal{O}(\texttt{seqlen})$ while comms $\sim \mathcal{O}(1)$ at fixed `cp_degree`. Right: strong scaling at fixed `seqlen` and varying `cp_degree` for the compute-then-correct strategy. Per-GPU performance is relatively stable across GPU-count and `seqlen`.

### References

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[5] NVIDIA *et al.*, "Nemotron-H: A Family of Accurate and Efficient Hybrid Mamba-Transformer Models." [Online]. Available: https://arxiv.org/abs/2504.03624

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[7] "causal-conv1d." [Online]. Available: https://github.com/Dao-AILab/causal-conv1d